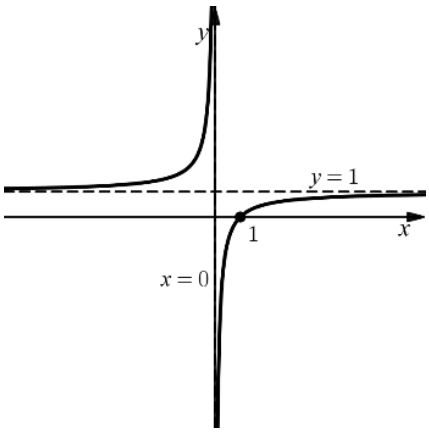
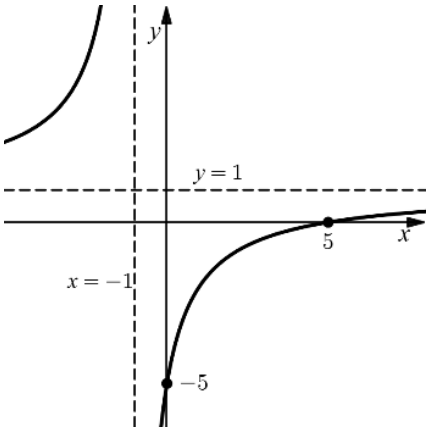
1.18 Graph Sketching: Linear Rational Functions

Worked Solutions

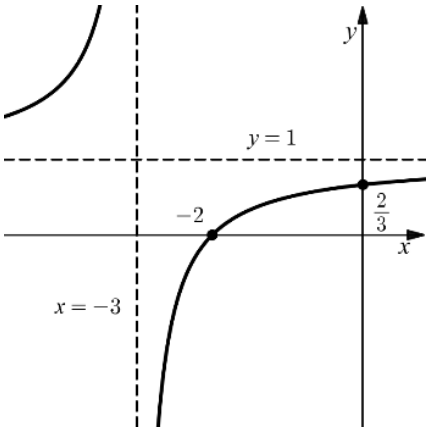
1 a)
When $y = 0$, $x = 1$
Vertical Asymptote: $x = 0$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



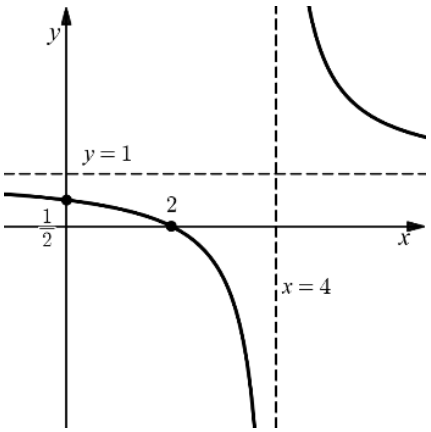
1 b)
When $x = 0$, $y = -5$, when $y = 0$, $x = 5$
Vertical Asymptote: $x = -1$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



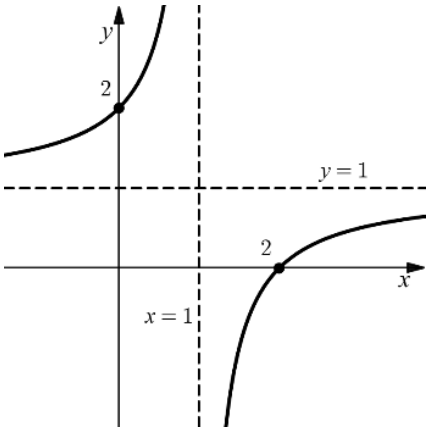
1 c)
When $x = 0$, $y = \frac{2}{3}$, when $y = 0$, $x = -2$
Vertical Asymptote: $x = -3$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



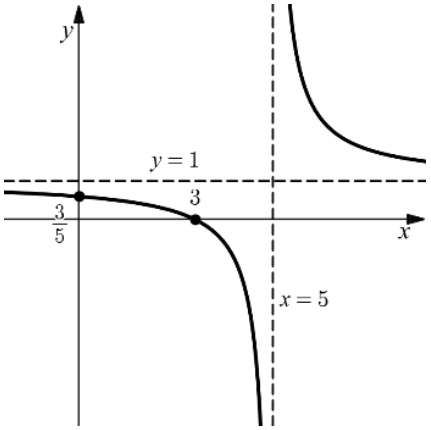
1 d)
When $x = 0$, $y = \frac{2}{3}$, when $y = 0$, $x = -2$
Vertical Asymptote: $x = -3$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



1 e)
When $x = 0$, $y = 2$, when $y = 0$, $x = 2$
Vertical Asymptote: $x = 1$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



1 f)
When $x = 0$, $y = 1$, when $y = 0$, $x = 3$
Vertical Asymptote: $x = 5$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



2 a)
Vertical asymptote gives denominator of $x - 2$
When $y = 0$, $x = -1$

$$y = \frac{x+1}{x-2}$$

2 b)
Vertical asymptote gives denominator of $x - 1$
When $y = 0$, $x = 3$

$$y = \frac{x-3}{x-1}$$

2 c)
Vertical asymptote gives denominator of $x+1$
When $y=0$, $x=-2$

$$y=\frac{x+2}{x+1}$$

2 d)
Vertical asymptote gives denominator of $x+2$
When $y=0$, $x=3$

$$y=\frac{x-3}{x+2}$$

3 a)
$y=3$, $x=2$

3 b)
$y=2$, $x=-\frac{1}{2}$

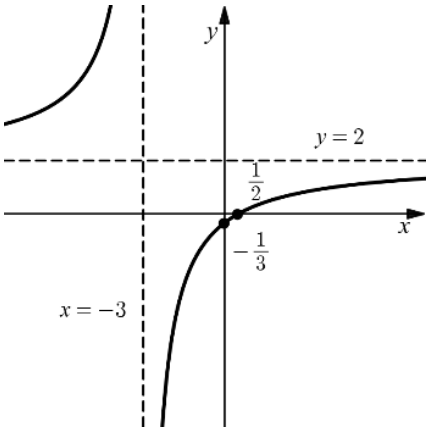
3 c)
$y=5$, $x=-2$

3 d)
$y=\frac{2}{3}$, $x=-\frac{5}{3}$

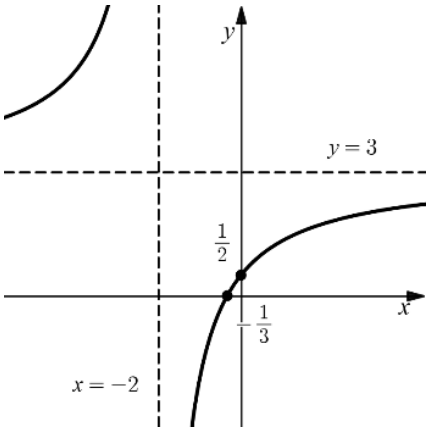
3 e)
$y=\frac{1}{5}$, $x=\frac{1}{5}$

3 f)
$y=\frac{a}{c}$, $x=-\frac{d}{c}$

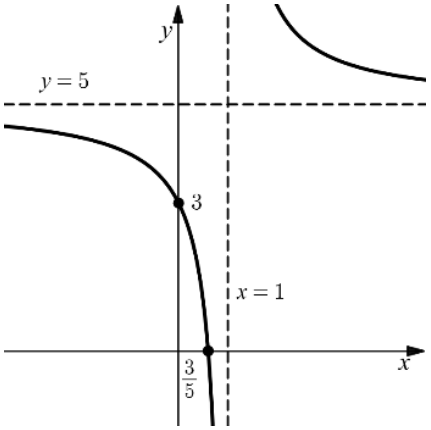
4 a)
When $x=0$, $y=-\frac{1}{3}$, when $y=0$, $x=\frac{1}{2}$
Vertical Asymptote: $x=-3$,
As $x\to\infty$, $y\to 2$ and as $x\to-\infty$, $y\to 2$



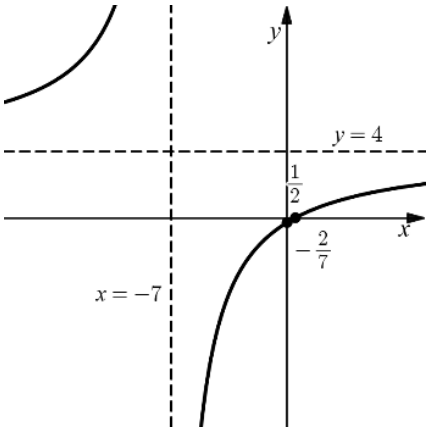
4 b)
When $x = 0$, $y = \frac{1}{2}$, when $y = 0$, $x = -\frac{1}{3}$
Vertical Asymptote: $x = -2$,
As $x \to \infty$, $y \to 3$ and as $x \to -\infty$, $y \to 3$



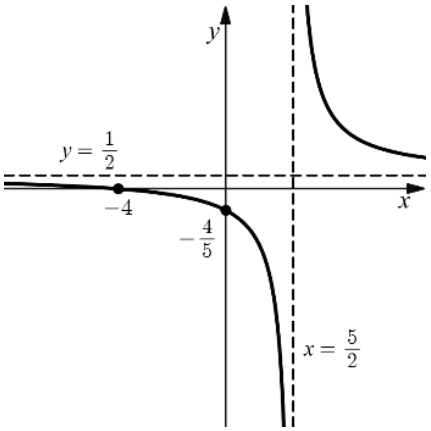
4 c)
When $x = 0$, $y = 3$, when $y = 0$, $x = \frac{3}{5}$
Vertical Asymptote: $x = 1$,
As $x \to \infty$, $y \to 5$ and as $x \to -\infty$, $y \to 5$



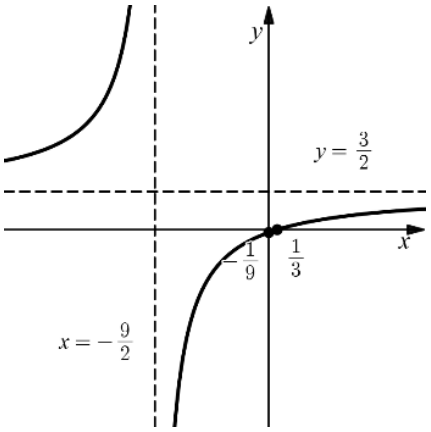
4 d)
When $x = 0$, $y = -\frac{2}{7}$, when $y = 0$, $x = \frac{1}{2}$
Vertical Asymptote: $x = -7$,
As $x \to \infty$, $y \to 4$ and as $x \to -\infty$, $y \to 4$



4 e)
When $x = 0$, $y = -\frac{4}{5}$, when $y = 0$, $x = -4$
Vertical Asymptote: $x = \frac{5}{2}$,
As $x \to \infty$, $y \to \frac{1}{2}$ and as $x \to -\infty$, $y \to \frac{1}{2}$



4 f)
When $x = 0$, $y = -\frac{1}{9}$, when $y = 0$, $x = \frac{1}{3}$
Vertical Asymptote: $x = -\frac{9}{2}$,
As $x \to \infty$, $y \to \frac{3}{2}$ and as $x \to -\infty$, $y \to \frac{3}{2}$



5 a)
Vertical asymptote gives denominator of $x - 1$
When $y = 0$, $x = \frac{1}{2}$
Horizontal asymptote gives ratio of x-terms as 2

$$y = \frac{2x - 1}{x - 1}$$

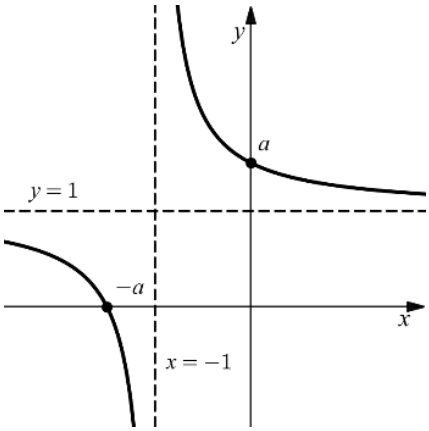
5 b)
Vertical asymptote gives denominator of $x + 2$
When $y = 0$, $x = -1$
Horizontal asymptote gives ratio of x-terms as 3

$$y = \frac{3x + 3}{x + 2}$$

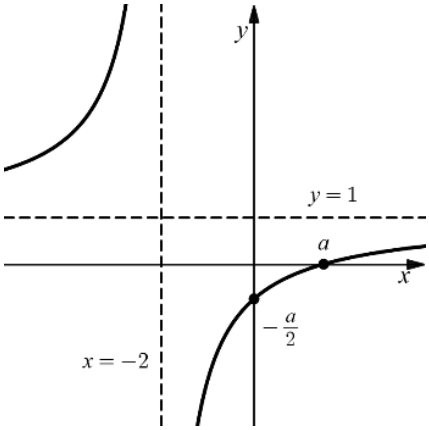
5 c)
Vertical asymptote gives denominator of $2x - 1$
When $y = 0$, $x = -\frac{1}{2}$
Horizontal asymptote gives ratio of x-terms as 3

$$y = \frac{6x + 3}{2x - 1}$$

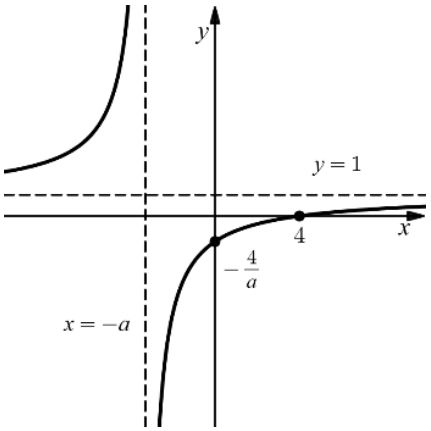
6 a)
When $x = 0$, $y = a$, when $y = 0$, $x = -a$
Vertical Asymptote: $x = -1$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



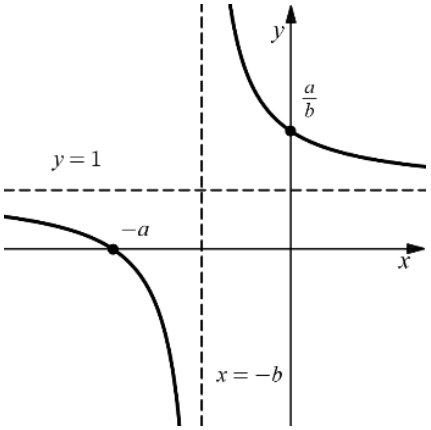
6 b)
When $x = 0$, $y = -\frac{a}{2}$, when $y = 0$, $x = a$
Vertical Asymptote: $x = -2$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



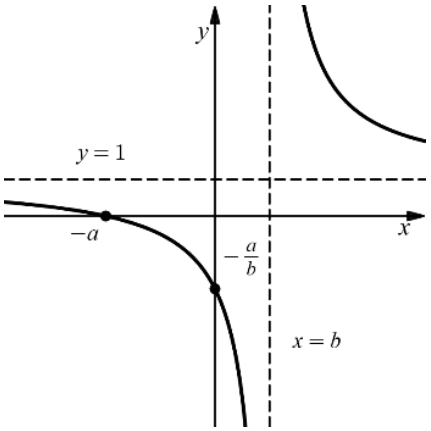
6 c)
When $x = 0$, $y = -\frac{4}{a}$, when $y = 0$, $x = 4$
Vertical Asymptote: $x = -a$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



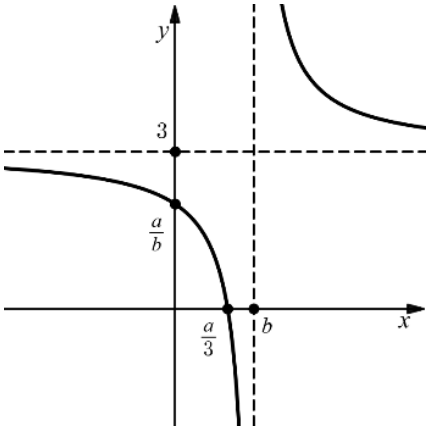
6 d)
When $x = 0$, $y = \frac{a}{b}$, when $y = 0$, $x = -a$
Vertical Asymptote: $x = -b$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



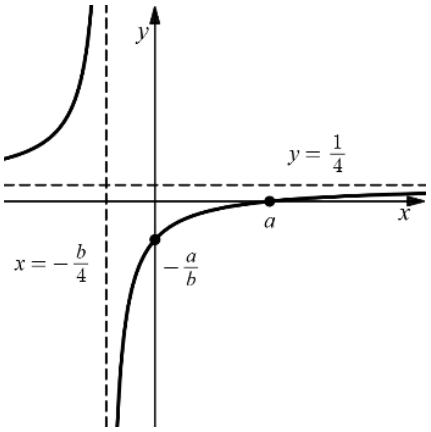
6 e)
When $x = 0$, $y = -\frac{a}{b}$, when $y = 0$, $x = -a$
Vertical Asymptote: $x = b$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$



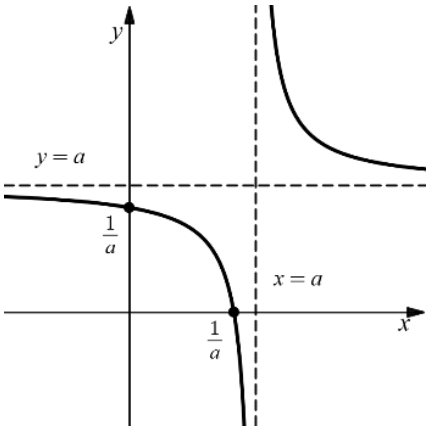
6 f)
When $x = 0$, $y = \frac{a}{b}$, when $y = 0$, $x = \frac{a}{3}$
Vertical Asymptote: $x = b$,
As $x \to \infty$, $y \to 3$ and as $x \to -\infty$, $y \to 3$



6 g)
When $x = 0$, $y = -\frac{a}{b}$, when $y = 0$, $x = a$
Vertical Asymptote: $x = -\frac{b}{4}$,
As $x \to \infty$, $y \to \frac{1}{4}$ and as $x \to -\infty$, $y \to \frac{1}{4}$

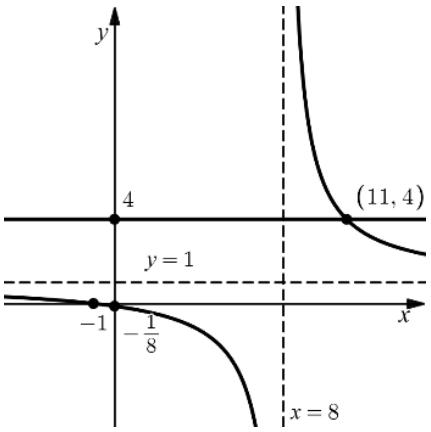


6 h)
When $x = 0$, $y = \frac{1}{a}$, when $y = 0$, $x = \frac{1}{a}$
Vertical Asymptote: $x = a$,
As $x \to \infty$, $y \to a$ and as $x \to -\infty$, $y \to a$



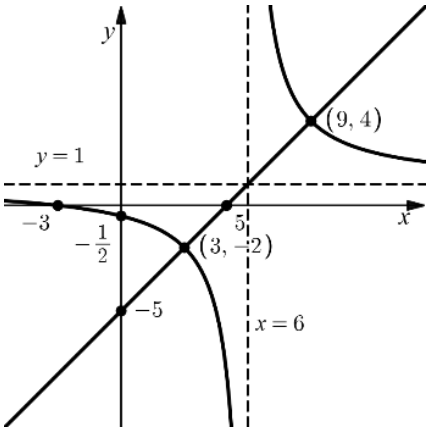
7 a)
When $x = 0$, $y = -\frac{1}{8}$, when $y = 0$, $x = -1$
Vertical Asymptote: $x = 8$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

$$4 = \frac{x+1}{x-8} \Rightarrow 4x - 32 = x + 1 \Rightarrow x = 11 \text{ so } (11, 4)$$



7 b)
When $x = 0$, $y = -\frac{1}{2}$, when $y = 0$, $x = -3$
Vertical Asymptote: $x = 6$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

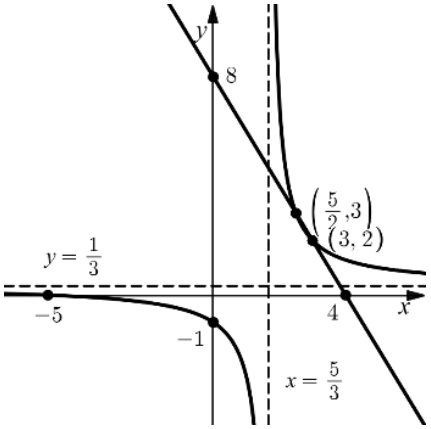
$$x - 5 = \frac{x+3}{x-6} \Rightarrow x^2 - 11x + 30 = x + 3$$
$$\Rightarrow x^2 - 12x + 27 = 0 \Rightarrow x = 3 \text{ or } x = 9 \text{ so } (3, -2) \text{ and } (9, 4)$$



7 c)
When $x = 0$, $y = -1$, when $y = 0$, $x = -5$
Vertical Asymptote: $x = \frac{5}{3}$,
As $x \to \infty$, $y \to \frac{1}{3}$ and as $x \to -\infty$, $y \to \frac{1}{3}$

$$8 - 2x = \frac{x+5}{3x-5} \Rightarrow -6x^2 + 34x - 40 = x + 5$$

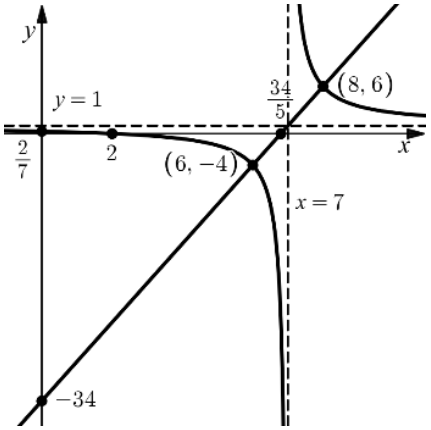
$$\Rightarrow 6x^2 - 33x + 45 = 0 \Rightarrow x = \frac{5}{2} \text{ or } x = 3 \text{ so } (\frac{5}{2}, 3) \text{ and } (3, 2)$$



7 d)
When $x = 0$, $y = 2$, when $y = 0$, $x = 2$
Vertical Asymptote: $x = 7$,
As $x \to \infty$, $y \to 1$ and as $x \to -\infty$, $y \to 1$

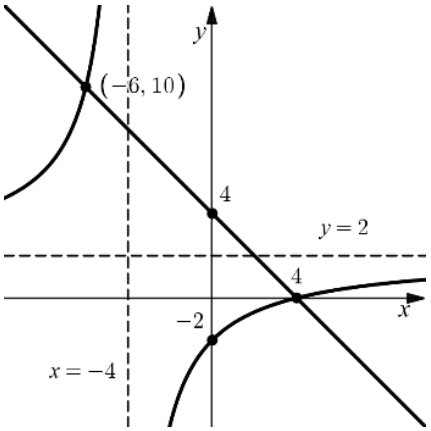
$$5x - 34 = \frac{x-2}{x-7} \Rightarrow 5x^2 - 69x + 238 = x - 2$$

$$\Rightarrow 5x^2 - 70x + 240 = 0 \Rightarrow x = 6 \text{ or } x = 8 \text{ so } (6, -4) \text{ and } (8, 6)$$



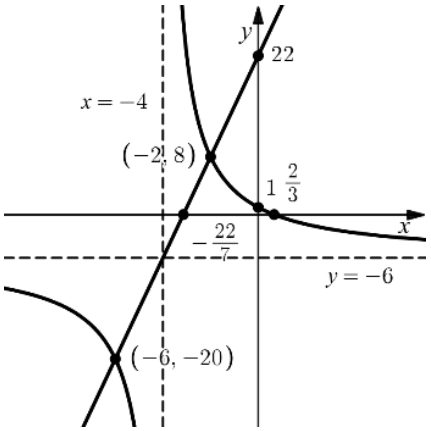
7 e)
When $x = 0$, $y = -2$, when $y = 0$, $x = 4$
Vertical Asymptote: $x = -4$,
As $x \to \infty$, $y \to 2$ and as $x \to -\infty$, $y \to 2$

$$4 - x = \frac{2x - 8}{x + 4} \Rightarrow -x^2 + 16 = 2x - 8$$
$$\Rightarrow x^2 + 2x - 24 = 0 \Rightarrow x = -6 \text{ or } x = 4 \text{ so } (-6, 10) \text{ and } (4, 0)$$



7 f)
When $x = 0$, $y = 1$, when $y = 0$, $x = \frac{2}{3}$
Vertical Asymptote: $x = -4$,
As $x \to \infty$, $y \to -6$ and as $x \to -\infty$, $y \to -6$

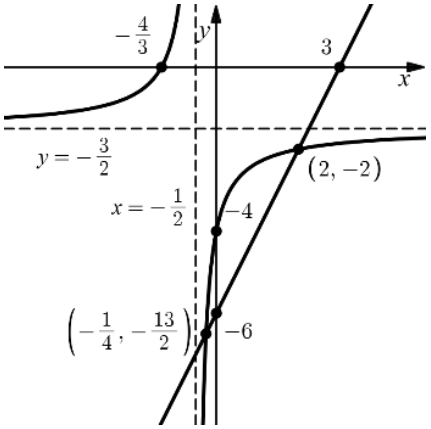
$$7x + 22 = \frac{4 - 6x}{x + 4} \Rightarrow 7x^2 + 50x + 88 = 4 - 6x$$
$$\Rightarrow 7x^2 + 56x + 84 = 0 \Rightarrow x = -6 \text{ or } x = -2 \text{ so } (-6, -20) \text{ and } (-2, 8)$$



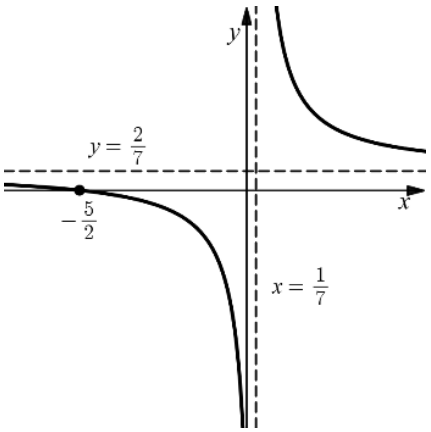
7 g)
When $x = 0$, $y = -\frac{3}{2}$, when $y = 0$, $x = -\frac{4}{3}$
Vertical Asymptote: $x = -\frac{3}{2}$,
As $x \to \infty$, $y \to -\frac{3}{2}$ and as $x \to -\infty$, $y \to -\frac{3}{2}$

$$2x - 6 = -\frac{3x+4}{2x+1} \Rightarrow 4x^2 - 10x - 6 = -3x - 4$$

$$\Rightarrow 4x^2 - 7x - 2 = 0 \Rightarrow x = 2 \text{ or } x = \frac{1}{4} \text{ so } (2, -2) \text{ and } (\frac{1}{4}, -\frac{13}{2})$$



8 a)
Asymptotes: $x = \frac{1}{7}$, $y = \frac{2}{7}$
Intercepts: $(0, -5)$, $(-\frac{5}{2}, 0)$



8 b)

$$y + 2 = \frac{2(x-1)+5}{7(x-1)-1} = \frac{2x+3}{7x-8}$$

$$y = \frac{2x+3}{7x-8} - 2 = \frac{2x+3}{7x-8} - 2 \times \frac{7x-8}{7x-8}$$
$$= \frac{2x+3-14x16}{7x-8} = \frac{-12x+19}{7x-8}$$

8 c)

$$\frac{-12(\frac{x}{2})+19}{7(\frac{x}{2})-8}=\frac{-6x+19}{\frac{7}{2}x-8}=\frac{-12x+38}{7x-16}$$

9 a)

$$\mathrm{f}(x)=\frac{3x-k}{x+4}$$

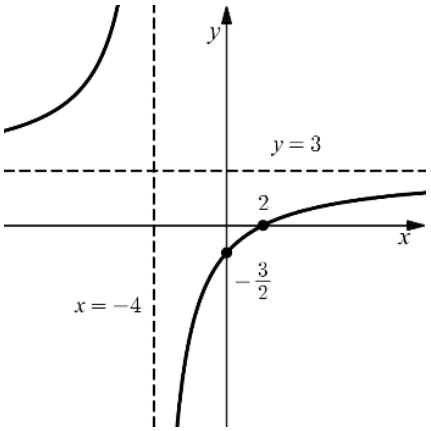
At $(2,0)$ then $0=6-k$ so $k=6$

$$\mathrm{f}(x)=\frac{3x-6}{x+4}$$

9 b)

When $x=0$, $y=-\frac{3}{2}$

9 c)



9 d)

$$\frac{3x-6}{x+4}=6$$
$$3x-6=6(x+4)$$
$$3x-6=6x+24$$
$$-30=3x$$
$$\therefore x=-10$$

10 a)

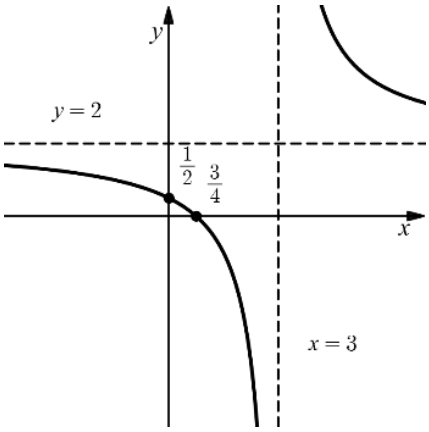
$$\mathrm{f}(x)=\frac{2x-k}{x-3}$$

At $(\frac{3}{4},0)$ then $0=2\times\frac{3}{4}-k$ so $k=\frac{3}{2}$

$$\mathrm{f}(x)=\frac{2x-\frac{3}{2}}{x-3}=\frac{4x-3}{2x-6}$$

10 b)
When $x = 0$, $y = \frac{1}{2}$

10 c)



10 d)

$$\frac{4x-3}{2x-6} = x + \frac{1}{2}$$
$$4x - 3 = (2x-6)(x+\frac{1}{2})$$
$$4x - 3 = 2x^2 - 5x - 3$$
$$2x^2 - 9x = 0$$
$$x(2x-9) = 0$$
$$x = 0 \text{ or } x = \frac{9}{2}$$
$$\therefore (0, \frac{1}{2}) \text{ and } (\frac{9}{2}, 5)$$

11 a)
$a = 5$, $b = 1$

11 b)

$$y = -\frac{1}{2}x$$

11 c)

$$-\frac{x}{2} = \frac{2x}{5x+1} \Rightarrow 5x^2 + x = -4x \Rightarrow 5x^2 + 5x = 0 \Rightarrow 5(x+1) = 0$$
$$\therefore (0,0) \text{ and } (-1, \frac{1}{2})$$

12 a)
$a = 7,\ b = 5$

12 b)

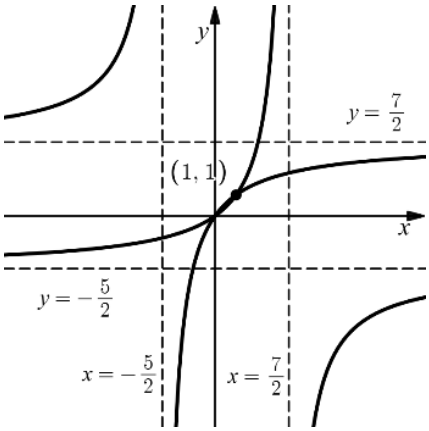
$$\frac{7 \times 1}{2 \times 1 + 5} = 1 \ \therefore (1, 1) \text{ lies on } C_1$$

12 c)

$$y = -\frac{5x}{2x - 7}$$

$$-\frac{5 \times 1}{2 \times 1 - 7} = -\frac{5}{-5} = 1 \ \therefore (1, 1) \text{ lies on } C_2$$

12 d)

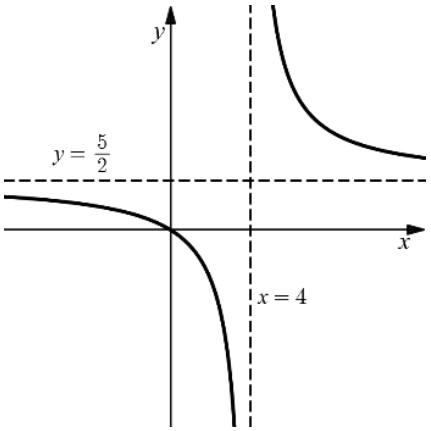


12 e)
The curves are reflection of each other in the line $y = x$

13 a)
$a = 5,\ b = 2,\ c = 8$

13 b)
Because the horizontal asymptote depends solely on the ratio $\frac{a}{b}$. If this ratio is not in simplest form, we could get multiple different values for a and b that all reduce to the same $\frac{a}{b}$, making the system underdetermined. By fixing $\frac{a}{b}$ in simplest form, we ensure a unique solution for all constants.

13 c)



13 d)

The one point of intersection of the curve with the axes is at the origin.

The line $y = -\frac{x}{8}$ passes exactly through this point and has a gradient which is the same as the curve.

The curve and the line meet only once, i.e they have exactly one intersection so that there is only one solution to the equation.